

Municipality Typologies and Data-Driven Revitalization Recommendations Using Socio-Economic Indicators and Regional News Signals

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Abstract

We propose a data-driven framework for place-based territorial revitalization that accounts for structural heterogeneity across municipalities. Using 2022 data for 2,006 Russian municipalities, we construct a two-stage typology in which municipalities are first grouped by demographic structure, measured by six gender–age population shares, and then clustered by socio-economic conditions using 18 selected indicators. This procedure yields nine interpretable municipality profiles and restricts comparisons to structurally similar territories. Building on this typology, we generate numeric development recommendations via cluster-constrained k-nearest neighbors (kNN): for each municipality, we identify similar peers within the same cluster and rank socio-economic indicators according to the largest gaps between the municipality and the mean values of its neighbors. To complement these structural recommendations with contextual signals, we incorporate regional news headlines from 2022. We apply semantic filtering based on anchor texts, topic modeling with BERTopic, and zero-shot classification into three policy-relevant categories: progress, neutral, and crisis. We then derive topic-specific "growth gap" and "pressure" scores by comparing a municipality's regional topic shares with those of the regions associated with its neighboring municipalities, weighted by estimated news intensity per capita. The framework produces combined numeric and topic-based recommendations and supports case-based interpretation for municipal development planning. We discuss limitations related to the regional aggregation of news, the quality of zero-shot labeling, and the absence of longitudinal outcome data.

Keywords: place-based policy; municipal clustering; kNN recommendations; topic modeling; BERTopic; zero-shot classification; regional development; migration proxy

1. Introduction

Regional and municipal development policies increasingly emphasize place-based interventions rather than one-size-fits-all solutions [1,2]. Large countries exhibit substantial heterogeneity across territories: metropolitan areas, industrial towns, agricultural districts, and remote municipalities often differ in demographic structure, economic capacity, infrastructure, and public service provision. As a result, direct comparisons between very different territories may lead to misleading policy conclusions and poorly targeted revitalization measures. This paper introduces a practical machine-learning framework that aims to support more targeted territorial revitalization by (i) identifying groups of comparable municipalities and (ii) providing data-driven recommendations derived from

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peer comparisons. We focus on municipalities as the primary unit of analysis, using a proxy outcome related to territorial attractiveness: net migration rate (net inflow minus outflow normalized by population). The central idea is that municipalities should be evaluated relative to structurally similar peers [3], not relative to national averages or fundamentally different territories. Our approach combines structured municipal indicators with contextual signals extracted from regional news. First, we build an interpretable municipality typology via two-stage clustering: demographic grouping based on gender–age population shares, followed by socio-economic clustering within each demographic group. Second, we generate numeric recommendations using cluster-constrained kNN: for each municipality, we identify similar neighbors in the feature space and rank the indicators where the municipality most lags behind its peers. Third, because municipal-level news data are typically unavailable, we incorporate regional news headlines as a contextual layer. We apply semantic filtering, topic modeling, and zero-shot classification to derive policy-relevant topic signals and compare these signals between a municipality’s region and the regions of its peer municipalities. The contributions of this paper are threefold: (1) a two-stage municipality typology that operationalizes "comparability" via demographic and socio-economic profiles; (2) a transparent recommendation mechanism based on peer gaps within clusters; (3) an integrated text-based layer that transforms regional news into topic-level gap and pressure indicators to complement numeric recommendations. Finally, we discuss validation strategies and limitations, including regional aggregation of news signals, moderate classification accuracy of zero-shot labeling, and the absence of longitudinal outcomes in the current dataset. Figure 1 provides an overview of the full workflow. Although the framework is developed for municipal administrative units rather than individual cities, it is directly relevant to smart-city research: it demonstrates how heterogeneous structured indicators and unstructured news data can be combined into a transparent, place-aware decision-support tool — a recurring concern in data-driven urban governance.

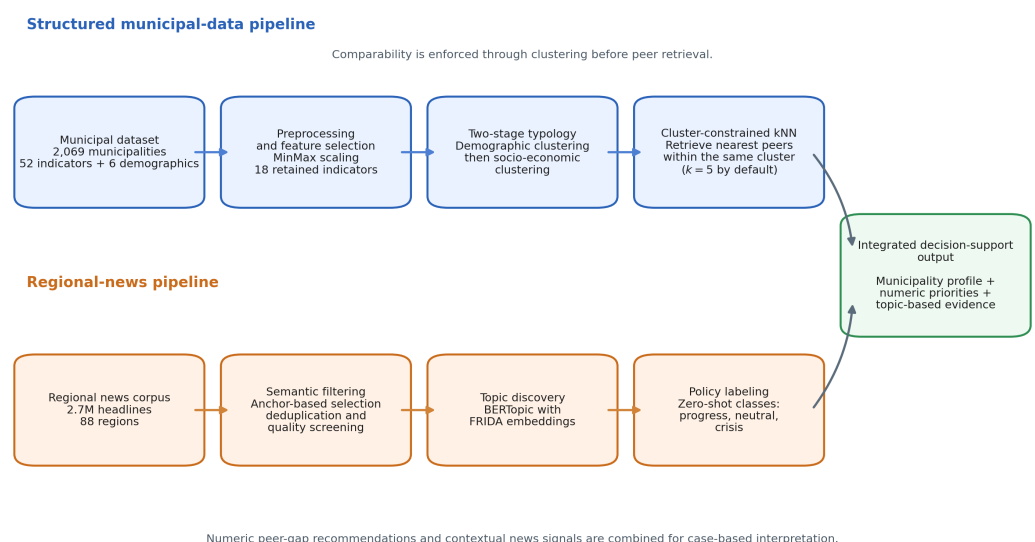


Figure 1. Overview of the proposed analytical pipeline. Structured municipal data are transformed into a two-stage typology and cluster-constrained numeric recommendations, while regional news headlines are filtered, organized into topics, and labeled to generate contextual signals. The two information streams are then combined into the final municipality-level decision-support output.

2. Related Work

2.1. Place-based regional development policy

Place-based development policy argues that effective territorial interventions must be tailored to local constraints and strengths rather than applied uniformly across heterogeneous territories [1]. In large and spatially diverse countries, one-size-fits-all programs risk misallocating resources: metropolitan areas, industrial towns, agricultural districts, and remote settlements differ in demographic structure, economic capacity, and public service provision, and the returns to a given intervention vary accordingly. Rodríguez-Pose [2] further argues that the neglect of such “places that don’t matter” generates not only economic inefficiency but also political backlash, reinforcing the case for explicitly territorial policy design. Recent empirical work on regional inequality in Europe [3] formalizes this intuition through the notion of “regional clubs”: groups of regions that share developmental trajectories and for which different bundles of interventions are appropriate. Taken together, this literature motivates two practical requirements for data-driven decision support in territorial policy — (i) the construction of comparable territorial groups, so that benchmarking is defensible, and (ii) recommendations that reflect the specific developmental position of each group rather than national averages. These requirements are particularly salient for Russia, where interregional and intraregional inequality has remained high throughout the post-Soviet period [4]. Population concentration has intensified in only a small fraction of the national territory [5], a pattern that reflects systemic shrinkage of the Russian urban system as a whole [6]. Shrinkage of small and medium-sized cities is widespread and largely under-addressed in formal spatial planning [7]. Recent trajectories of Russian shrinking cities exhibit systematic patterns that can be captured through typological analysis [8], suggesting that municipality-level classification is a useful starting point for targeted revitalization.

2.2. Territorial typologies: from official classifications to data-driven clustering

Statistical agencies have long produced descriptive territorial typologies to support comparable analysis across units. The Eurostat methodological manual [9] operationalizes territorial heterogeneity at the NUTS and LAU levels primarily through population density and urbanization thresholds, yielding classifications such as predominantly urban, intermediate, and predominantly rural regions. While useful as a common reference frame, such typologies are intentionally coarse and are not designed to capture the multidimensional socio-economic conditions that drive local development outcomes. Data-driven clustering offers a complementary approach that can scale to thousands of municipalities while remaining interpretable when cluster profiles are summarized by centroids and a small set of key indicators. Early applications demonstrated the feasibility of k-means clustering on municipal socio-economic indicators for policy targeting [10], and subsequent country-scale studies have applied k-means to hundreds of variables across national municipal systems [11,12]. More recent work has moved toward richer unsupervised pipelines: Tonini et al. [13] redefine Swiss territorial typologies on roughly 2,172 municipalities using self-organizing maps and hierarchical clustering over physical, demographic, and socio-economic variables, producing interpretable rural–urban continuum classes. This line of research establishes unsupervised learning on municipal-level socio-economic data as an accepted methodological approach, and provides a natural international benchmark for the present study. Within the Russian context, Tarasieva et al. [14] formulate the optimization of a socio-economic system aimed at increasing residents’ satisfaction and combine econometric modeling with machine-learning-based analysis, illustrating the broader shift toward ML-augmented municipal decision support. Our contribution extends this line of work along two axes. First, existing data-driven typologies typically cluster

municipalities in a single stage over a unified feature set, whereas our two-stage design first partitions municipalities by demographic structure and then clusters within each demographic group by socio-economic indicators. This reflects the view that demographic composition and socio-economic condition describe distinct facets of municipal structure and should not be mixed within a single distance metric. Second, we embed the resulting typology in a recommendation mechanism rather than treating it as a descriptive end product. Recommendation-style methods based on similarity search provide a transparent alternative to black-box optimization: a target unit is compared to structurally similar peers, and the largest feature gaps define a practical “what to improve” agenda. Contemporary work on k-nearest-neighbor recommenders has refined such designs around the choice of similarity function, neighborhood constraints, and aggregation rules [15]. We adopt a cluster-constrained variant in which the candidate peer set is restricted to the target municipality’s cluster, ensuring that recommended targets reflect feasible benchmarks achieved by comparable territories rather than global best-in-class values.

2.3. Text-as-data and NLP for policy measurement

A parallel strand of work has established the use of text-as-data for economic and policy measurement. Gentzkow et al. [16] survey this paradigm, showing how large corpora of news articles, public communications, and administrative text can be converted into structured indicators of local priorities, policy agendas, and perceived problems. A canonical empirical application is news-sentiment measurement: Shapiro et al. [17] demonstrate that NLP-derived news indices track macroeconomic conditions closely enough to be used by central-bank research units, providing direct evidence that policy-relevant signals can be recovered from news text even in the absence of explicit sentiment labels. Two methodological families are particularly relevant for extracting such signals at scale. The first is embedding-based topic discovery, which replaces bag-of-words representations with contextual sentence embeddings [19] and then clusters documents in the resulting semantic space. BERTopic [20] is a widely used instance of this family; it combines transformer embeddings, UMAP-based dimensionality reduction [22], HDBSCAN density-based clustering [23], and a class-based TF-IDF step for topic representation. Comparative evaluations on short texts consistently find BERTopic competitive with or superior to classical LDA- and NMF-based approaches, particularly for headline-length documents [24], which motivates its use in the present study. Recent applications of topic modeling in smart-city contexts [18] further illustrate how thematic analysis of textual data can surface urban governance themes that complement structural indicators. The second family is zero-shot classification based on natural-language inference, introduced by Yin et al. [25], which reformulates classification as entailment between a document and a set of label hypotheses. This approach allows documents to be mapped into custom policy-relevant categories — such as progress, neutral, and crisis — without labeled training data, at the cost of moderate accuracy and potential systematic biases. For Russian-language corpora, the practical feasibility of zero-shot classification rests on multilingual representation learning; the XLM-R model of Conneau et al. [26] provides the cross-lingual backbone that underpins the Russian-compatible NLI models used in this study. In our setting, municipal-level news coverage is unavailable and regional aggregates are the finest granularity we can obtain reliably. We therefore use regional headlines as a contextual layer that complements, rather than replaces, municipal structural indicators. To our knowledge, the integration of cluster-constrained peer-gap recommendations with region-level topic signals derived through BERTopic and zero-shot NLI has not been systematically explored for Russian municipalities, and this combination constitutes the principal methodological contribution of the present paper.

3. Data

The municipal dataset for 2022 is drawn from the official statistics of the Russian Federation¹. It covers 2,006 municipalities and 52 raw socio-economic indicators, of which 18 are retained after feature selection (Section 4.2). Table 1 lists these indicators together with their descriptions, units, and the notation used in the remainder of the paper; most are reported per capita at source.

Table 1. Selected socio-economic indicators used for clustering and recommendation generation. The unit column indicates the original measurement scale and whether the indicator was expressed per capita in the source dataset.

Name	Description	Unit	Symbol
consnewareas	Newly commissioned residential housing area.	m ² per capita	x_{cna}
popsize	Total municipal population.	persons	x_{pop}
foodservturnover	Public catering turnover.	thousand RUB per capita	x_{fst}
roadslen	Length of public roads in municipal ownership.	km per capita	x_{rds}
socialaid	Number of social service centers for elderly people and persons with disabilities.	units per capita	x_{sad}
invest	Investment in fixed capital.	thousand RUB per capita	x_{inv}
factoriescap	Value of shipped own-produced goods.	thousand RUB per capita	x_{fac}
cultureorg	Number of cultural and leisure organizations.	units per capita	x_{cul}
livarea	Residential floor area per resident.	m ² per person	x_{liv}
sportsvenue	Number of sports facilities.	units per capita	x_{spt}
beforeschool	Number of places in preschool institutions.	places per capita	x_{pre}
goodcompincome	Financial result (profit) of organizations.	thousand RUB per capita	x_{gci}
hospitals	Number of medical and preventive care organizations.	units per capita	x_{hos}
retailturnover	Retail trade turnover excluding small enterprises.	thousand RUB per capita	x_{ret}
harvest	Crop yield.	centners per hectare	x_{hrv}
agrprod	Agricultural output.	thousand RUB per capita	x_{agr}
schoolnum	Number of general education organizations.	units per capita	x_{sch}
avgsalary	Average monthly salary.	RUB	x_{sal}

The same source provides population counts by age and gender, from which we construct six demographic shares defined by three age cohorts (children, working-age, elderly) and two genders (male, female). These shares describe population structure rather than economic conditions and are treated as a distinct feature group: they are the input to the first-stage demographic clustering, while the indicators in Table 1 feed the second-stage socio-economic clustering. In addition, for our analysis we use the migration balance for each municipality as an indicator of local attractiveness to residents. Migration data were likewise obtained from the official statistics.

As a proxy outcome for territorial attractiveness, we use the net migration rate — the balance of arrivals and departures relative to municipal population — which is also provided by the official statistics:

$$M(m) = \frac{\text{inflow}(m) - \text{outflow}(m)}{\text{pop}(m)} \quad (1)$$

where $\text{inflow}(m)$ and $\text{outflow}(m)$ are the numbers of residents moving into and out of municipality m during the reference year, and $\text{pop}(m)$ is its total population. Higher values of $M(m)$ indicate that the municipality attracts more residents than it loses.

Regional news headlines for 2022 were automatically collected from the open-access news aggregator archive². The resulting corpus contains approximately 2.7 million headlines aggregated across 88 regions. The preprocessing pipeline includes semantic filtering based on anchor texts, duplicate removal, a custom stopword list to exclude routine service announcements, and the removal of interrogative and very short headlines (fewer than five words). After filtering, 452,126 headlines remained for topic modeling.

¹ <https://rosstat.gov.ru>

² <https://bezformata.com>

4. Methodology

4.1. Two-stage clustering for municipality typology

We apply a two-stage clustering strategy to avoid comparing structurally dissimilar territories.

Stage 1 (demographic clustering). Using the demographic data, we calculate six gender–age population shares according to Equation (2):

$$s_{i,j} = \frac{p_{i,j}}{\mathcal{P}}, \quad i \in \{\text{children, working-age, elderly}\}, \quad j \in \{\text{male, female}\}, \quad (2)$$

where $s_{i,j}$ is the share of a certain cohort, $p_{i,j}$ is the number of the cohort representatives in the municipality and $\mathcal{P}$ is the total population size of this municipality. Municipalities are then clustered into three demographic groups using a Gaussian Mixture Model (GMM) with three components. GMM was preferred over k-means at this stage because it produces a comparable silhouette score while yielding more balanced cluster sizes — in particular, it isolates younger municipalities as a coherent profile rather than absorbing them into a larger group. Figure 2 shows the demographic centroids of these groups based on the six gender–age population shares, highlighting the contrast between younger, middle-aged, and older municipality profiles.

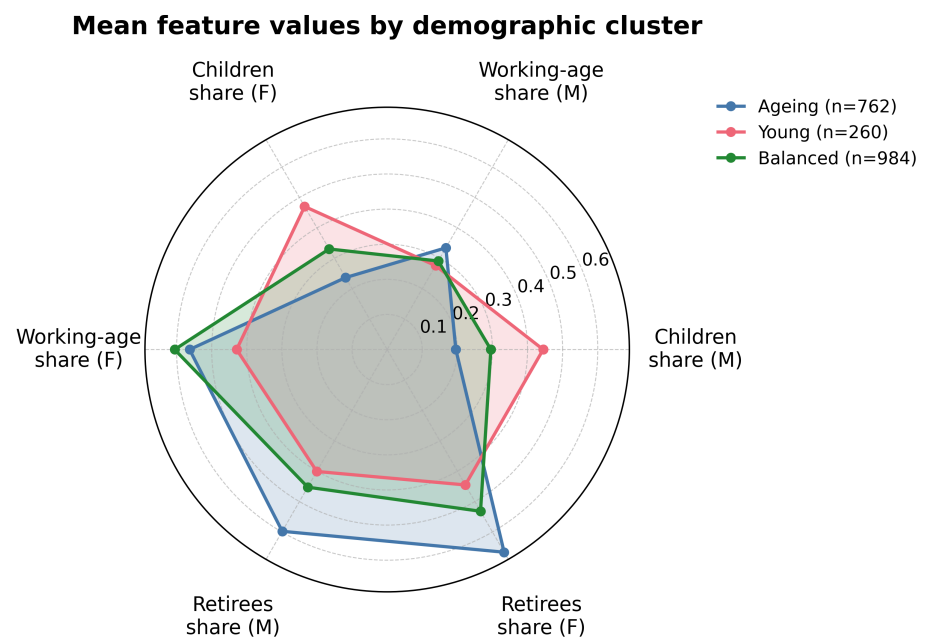


Figure 2. Radar chart of the demographic cluster centroids based on six gender–age population shares. The three groups capture municipalities with relatively younger, middle-aged, and older population structures and provide the first-stage typology used for subsequent within-group socio-economic clustering.

Stage 2 (socio-economic clustering within demographic groups). Within each demographic group, we cluster municipalities using socio-economic indicators selected via the feature selection ensemble (Section 4.2). We use k-means [30] with 3 clusters inside each demographic group, resulting in $3 \times 3 = 9$ municipality types overall. The number of clusters is selected via an automated elbow criterion following Morgado et al. [31]. For the within-group socio-economic stage, k-means is preferred over GMM due to higher silhouette scores; GMM produces similar partitions.

4.2. Feature selection (ensemble)

To derive a compact and informative feature set for clustering and recommendations, we aggregate four complementary methods: (i) Random Forest regression feature importance [27] with net migration as the target, (ii) XGBoost regression feature importance [28], (iii) VarianceThreshold to remove near-constant indicators, and (iv) selection of the L best features based on mutual information [29] with respect to the target. The final socio-economic feature set contains the 18 indicators listed in Table 1.

4.3. Numeric recommendations via cluster-constrained kNN gaps

For a target municipality m , we restrict the candidate peer set to municipalities belonging to the same cluster, so that recommendations are derived only from structurally comparable territories. Within this subset, we identify the n nearest neighbors in the selected feature space using Euclidean distance on MinMax-scaled variables, with $n = 5$ by default (or fewer if the cluster contains too few municipalities). We then quantify, for each feature f , how far the target municipality falls below the average level observed among its nearest peers (3):

$$\Delta_f = \bar{f}_{\mathcal{N}(m)} - f(m), \quad (3)$$

where $\bar{f}_{\mathcal{N}(m)}$ denotes the mean value of feature f across the neighbor set $\mathcal{N}(m)$. Features with the largest positive Δ_f indicate the dimensions in which the municipality lags most strongly behind comparable peers and are therefore interpreted as the highest-priority areas for improvement. This procedure yields a transparent, rank-ordered recommendation list grounded in peer benchmarking.

4.4. Text pipeline: semantic filtering, topic modeling, and policy labeling

As municipal-level news data are unavailable, we extract contextual signals from regional news headlines. We first perform semantic filtering by computing text embeddings and retaining headlines close to anchor texts representing socio-economic development and quality-of-life themes (e.g., “socio-economic development”, “quality of life”, “labor market”, “healthcare and education”, etc.). We then apply BERTopic using the FRIDA embedding model by ai-forever³, selected as the top-performing model for Russian-language tasks on the MTEB / MMTEB leaderboard at the time of analysis [21], with minimal cluster size of 50 and n-grams (1,2). This yields 1,119 topics including an outlier cluster (-1). Next, each headline is labeled via zero-shot classification with the language model xlm-roberta-large-xnli by Joe Davison⁴ using the template “This news headline is related to the following {...}” and labels {“progress and achievements”, “neutral information”, “crisis and problems”}. Predictions with confidence below 0.5 are treated as noise.

4.5. Topic-based gaps: growth gap and pressure

For each region r , topic t , and label $\ell \in \{\text{progress, crisis}\}$, we define the regional topic share as (4)

$$\mathcal{T}(r, t, \ell) = \frac{N(r, t, \ell)}{N(r)}, \quad (4)$$

³ <https://huggingface.co/ai-forever/FRIDA>

⁴ <https://huggingface.co/joeddav/xlm-roberta-large-xnli>

where $N(r, t, \ell)$ is the number of headlines in topic t with label ℓ in region r , and $N(r)$ is the total number of headlines in region r . To approximate municipal exposure to regional news, we define

$$\mathcal{H}(m) = \frac{\text{News}(r(m)) \times \text{pop}(m)}{\text{Pop}(r(m))}, \quad (5)$$

where $r(m)$ denotes the region containing municipality m , $\text{News}(r(m))$ is the number of filtered headlines for that region, $\text{pop}(m)$ is the municipal population (as in Section 3), and $\text{Pop}(r(m))$ is the total population of the region. Intuitively, $\mathcal{H}(m)$ scales region-level news volume by the municipality's share of the regional population, approximating the portion of regional coverage attributable to m under a uniform-exposure assumption.

The target municipality signal is then given by

$$\mathcal{G}(m, t, \ell) = \mathcal{T}(r(m), t, \ell) \times \mathcal{H}(m). \quad (6)$$

Let S denote the set of kNN neighbor municipalities of m within the same cluster, and let $R(S)$ be the set of regions containing those neighbors. We define the neighborhood topic share and neighborhood weight as

$$d_{\text{share}}(t, \ell) = \frac{1}{|R(S)|} \sum_{r \in R(S)} \mathcal{T}(r, t, \ell), \quad (7)$$

$$d_{\text{weight}}(m) = \frac{1}{|S|} \sum_{m' \in S} \mathcal{H}(m'), \quad (8)$$

so that the neighborhood signal is

$$\mathcal{D}(m, t, \ell) = d_{\text{share}}(t, \ell) \cdot d_{\text{weight}}(m). \quad (9)$$

Finally, we define the topic-based recommendation scores as growth gap P_g (10) ("good news") and pressure P_b (11) ("bad news").

$$P_g(m, t) = \mathcal{D}(m, t, \text{progress}) - \mathcal{G}(m, t, \text{progress}), \quad (10)$$

$$P_b(m, t) = \mathcal{G}(m, t, \text{crisis}) - \mathcal{D}(m, t, \text{crisis}). \quad (11)$$

Topics with the largest growth gap indicate areas in which peer municipalities exhibit more "progress" coverage than the target municipality, whereas large pressure values indicate relatively higher "crisis" coverage. Top-ranked topics are used to generate qualitative recommendations supported by representative neighbor-region headlines filtered by model confidence.

5. Results

5.1. Two-stage municipality typology

Applying the two-stage clustering pipeline to 2,006 municipalities yields nine municipality types (3 demographic groups $\times$ 3 socio-economic clusters per group). The first stage groups municipalities by six gender-age population shares into three demographic profiles (e.g., younger, balanced, aging; labels can be finalized based on centroid patterns). The second stage, performed separately within each demographic profile, clusters municipalities using 18 socio-economic indicators selected by the feature-selection ensemble. We find that this hierarchical design improves comparability: municipalities are benchmarked against peers with similar demographic structure and socio-economic conditions, avoiding direct comparisons between fundamentally different territories.

Figure 3(a)–3(c) report the centroid profiles of the socio-economic clusters obtained within the three demographic groups. Taken together, these radar plots show how the cluster structure varies across municipalities with predominantly young, balanced, and ageing population compositions.

Cluster labels follow the two-stage typology: Y1–Y3 denote the three socio-economic sub-clusters within the Young demographic group, and analogously B1–B3 (Balanced) and A1–A3 (Ageing).

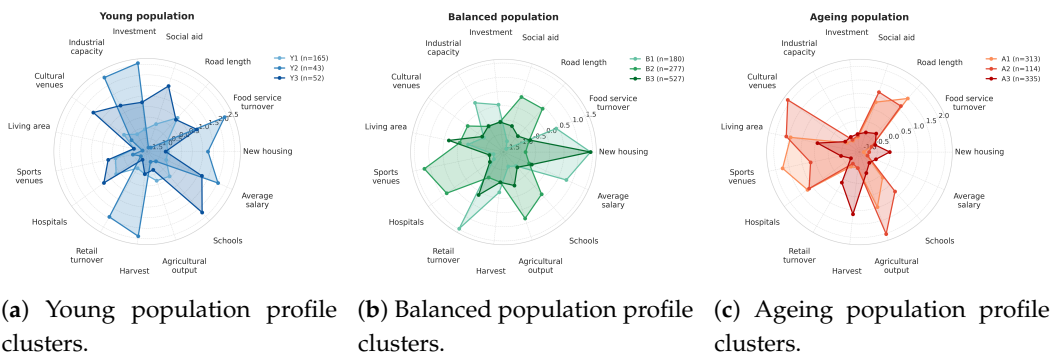


Figure 3. Radar plots of MinMax-scaled socio-economic cluster centroids within each demographic group. Each panel summarizes the characteristic indicator balances of the three clusters identified among municipalities with young, balanced, and ageing population structures, respectively.

Cluster centroids reveal systematic differences across the selected indicators, including income proxies (average salary, “good composite income”), infrastructure and amenities (road length, sports venues, cultural organizations), service activity (retail and food service turnover), and public service capacity (hospitals, schools, preschool facilities). The radar plots provide an interpretable fingerprint of each cluster and support qualitative naming, such as service-intensive and high-income, infrastructure-limited, or agro-oriented profiles.

Mapping cluster assignments onto Russian municipal geography reveals coherent spatial patterns (Figure 4). Young-population clusters concentrate in Siberia, the Far East, and the Arctic—territories whose demographic profile reflects historical patterns of industrial settlement and rotational labor migration documented in prior work [5,6]. Ageing-population clusters dominate European Russia, particularly the Central, Northwest, and Volga federal districts, consistent with long-documented demographic ageing of the country’s historical core. Balanced-population clusters form a recognizable southern belt across the agrarian regions of the North Caucasus and Lower Volga. The fact that the typology—derived purely from gender–age shares and socio-economic indicators, with no spatial information given to the algorithm—recovers these well-known macro-regional patterns provides external validity for the clustering and supports its use as a basis for peer-comparable benchmarking.

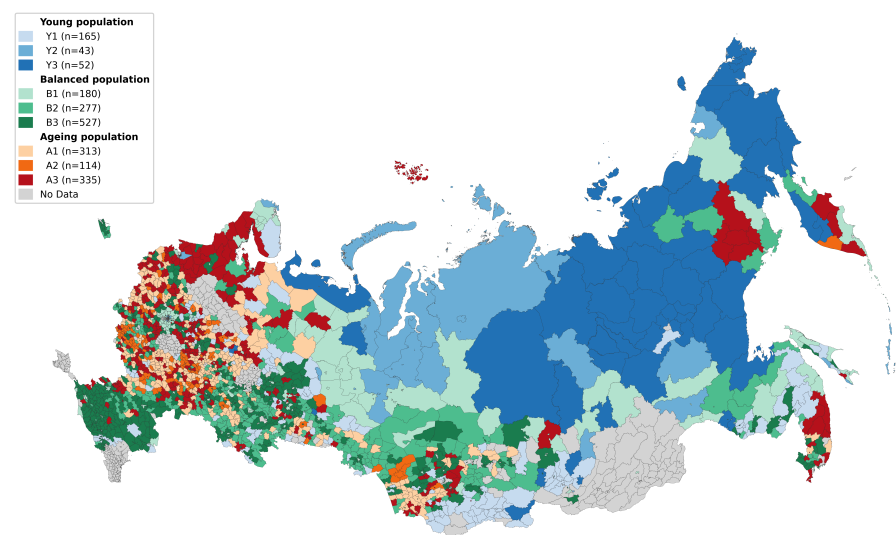


Figure 4. Spatial distribution of the demographic cluster assignments across Russian municipalities.

5.2. Numeric recommendations via cluster-constrained kNN

For each municipality, we retrieve up to $k = 5$ nearest neighbors within its final cluster using Euclidean distance on MinMax-scaled features. Recommendations are defined by feature gaps $\Delta_f = f_{\text{neighbors, mean}} - f(m)$, and we output the top-10 features with the largest positive gaps. This produces a transparent and actionable list of "priority factors" that the municipality lags behind relative to structurally comparable peers. Because neighbors are constrained to the same cluster, recommended targets reflect feasible peer benchmarks rather than national averages.

Figure 5 illustrates the numeric recommendation output for Kamensky Municipal District of cluster B3. The bars show the top-10 feature gaps Δ_f between the municipality and the mean values of its nearest in-cluster neighbors, so larger bars correspond to stronger improvement priorities under the peer-benchmarking scheme.

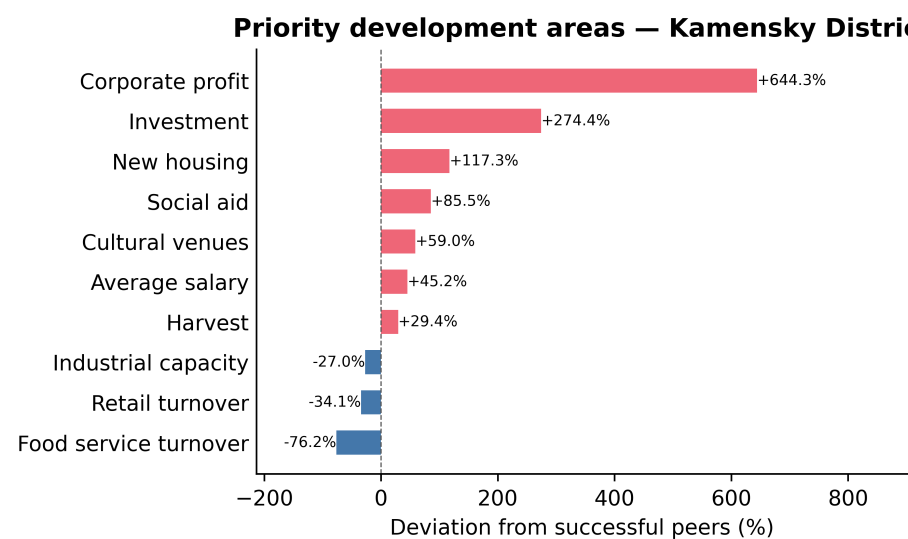


Figure 5. Example of numeric recommendations for Kamensky Municipal District. The chart ranks the top-10 socio-economic indicators for which the municipality lags behind the mean profile of its nearest comparable neighbors within the same cluster.

In this example, the recommended factors highlight the dimensions in which Kamensky Municipal District is most distant from its peer benchmark. The plot is intended

as a decision-support summary: it does not prescribe a policy instrument directly, but it identifies the socio-economic variables for which closing the observed gaps would bring the municipality closer to the profile of structurally comparable and better-performing neighbors.

5.3. News-driven topic signals and gap/pressure recommendations

From 2.7M regional news headlines (88 regions, 2022), semantic filtering using anchor texts yields 452,126 headlines relevant to socio-economic development and quality-of-life topics. BERTopic (ai-forever/FRIDA; min_cluster_size=50; ngram_range=(1,2)) discovers 1,119 topics including an outlier topic (-1), with 208,738 headlines assigned to the outlier. Topic quality metrics show topic coherence (c_v) = 0.387 and topic diversity = 0.656 on the full set; after removing the outlier topic (-1), topics with fewer than 300 headlines, and 24 topics manually flagged as substantively irrelevant, coherence improves to 0.425, suggesting higher semantic consistency in the retained subset.

Each headline is labeled using zero-shot classification (joeddav/xlm-roberta-large-xnli) into three policy categories: progress/neutral/crisis, with a confidence threshold of 0.5 (lower-confidence predictions treated as noise). On a manually labeled evaluation set ($n = 461$), performance is moderate: accuracy = 0.597, macro- F_1 = 0.592, and weighted- F_1 = 0.594; class-wise F_1 scores are 0.680 (progress), 0.537 (neutral), and 0.558 (crisis). Despite limitations, these labels enable a structured comparison of topic shares between a municipality's region and the regions of its peer municipalities.

We compute topic shares at the regional level and map them to municipalities using the estimated news-per-population weight. For each municipality, we compare its target signal with the neighborhood signal derived from its kNN peers' regions and compute two indicators:

growth_gap: where peers show more "progress" coverage than the target,

pressure: where the target shows more "crisis" coverage than peers.

In the default configuration, we report the top-3 topics by growth_gap and the top-3 topics by pressure, and attach representative neighbor-region headlines (filtered by model confidence) as qualitative evidence.

Table 2. Top-3 news-driven recommendation topics for Kamensky Municipal District. Growth-gap topics indicate areas in which peer regions show relatively stronger *progress* coverage; pressure topics indicate areas in which the target region shows relatively stronger *crisis* coverage.

#	Topic	Top keywords (BERTopic)	Representative headline
Growth-gap topics			
1	Housing renewal and new residential construction	renovations; residential complex; new microdistrict; quarter; construction	"A new comfort- and business-class residential district will be built in Omsk" (neighbor region)
2	Gubernatorial site inspections of public works	construction progress; inspected; inspected progress; checked progress; construction	"Outcomes of the Governor's public reception: new school, outpatient clinic, mobile feldsher-midwife station, road repairs, and stadium reconstruction" (neighbor region)
3	Housing rights enforcement for orphans	orphans; orphan children; prosecutorial intervention; parental care	"In Yekaterinburg, three orphans received well-appointed apartments in a new building following prosecutorial lawsuits" (neighbor region)
Pressure topics			
1	Flood preparedness and emergency response	flood; emergency regime; natural disaster; Transbaikalian districts; basements	"Viktor Tomenko: We continue preparing for the most challenging flood scenarios" (target region)
2	Accessibility and services for persons with disabilities	disabled persons; limited abilities; rehabilitation; for children; mobility-impaired	"Altai Krai discussed problems of the accessible environment" (target region)
3	Solid waste management reform	waste; handling; landfills; collection; Russian Environmental Operator	"Problems of waste management reform remain unresolved" (target region)

Read jointly with the numeric recommendations in Figure 5, the topic-based output (Table 2) adds qualitative texture rather than overlapping content. The numeric layer flags corporate profit, investment, and new housing as the largest peer gaps; the textual

layer surfaces the kinds of stories that peer regions are telling about housing — "new microdistricts", "comfort and business-class development" — and the kinds of stories that the target region is telling about itself — flood preparation, contested waste reform, accessibility shortfalls. The two layers complement each other: numeric gaps identify what to improve, topic gaps suggest how peer regions frame and communicate progress on similar dimensions. The Kamensky case also illustrates a notable convergence between the two pipelines: the largest numeric gaps include "new housing", while the top growth-gap topic is dominated by neighbor-region coverage of new residential construction. Because the numeric and textual signals are derived from independent data sources — official statistics and regional news — this agreement provides a degree of mutual cross-validation: both pipelines are picking up the same underlying developmental gap from different observational angles, rather than reproducing each other’s artifacts.

6. Validation and Robustness

We validate the proposed framework along three dimensions appropriate for a cross-sectional setting: the internal consistency and stability of the typology (Section 6.1), the robustness of the numeric recommendation mechanism (Section 6.2), and the reliability of the text-based signal pipeline (Section 6.3). Outcome-oriented validation against migration dynamics is left for future work, as it requires multi-year data that fall outside the scope of the present dataset.

6.1. Clustering validity and stability

We first assess internal cluster compactness using silhouette scores [32]. The first-stage demographic partition obtains a silhouette score of 0.241, while the within-group socio-economic partitions obtain silhouette scores of 0.316 (Young), 0.230 (Balanced), and 0.220 (Ageing). These values are moderate but consistent with what is typically reported for clustering on heterogeneous administrative data, where natural clusters are often partially overlapping. The slightly higher silhouette value in the Young demographic group suggests that this subset is more compact in feature space.

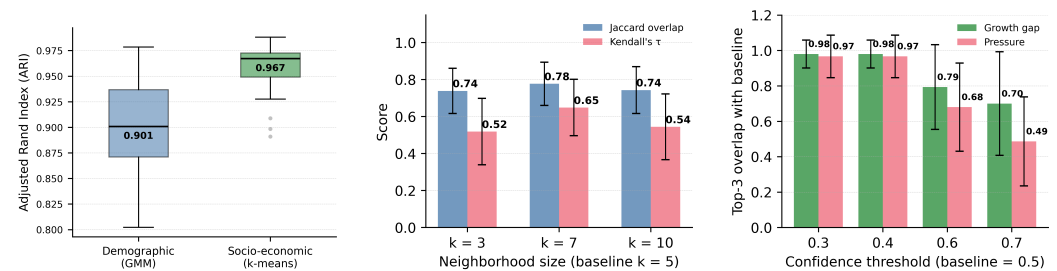
Cluster sizes range from 43 to 527 municipalities across the nine cells shown in Table 3. The smallest cell, Y_2 (the Young-group second sub-cluster) with $n = 43$, remains usable for kNN with $k = 5$, but recommendations within this cell rely on a smaller candidate peer pool and should therefore be interpreted with greater caution.

Table 3. Cluster size matrix for the two-stage typology. The notation Y_i , B_i , and A_i denotes socio-economic sub-clusters within the Young, Balanced, and Ageing demographic groups, respectively.

Demographic Group	Sub-cluster 1	Sub-cluster 2	Sub-cluster 3
Young	Y_1 ($n = 165$)	Y_2 ($n = 43$)	Y_3 ($n = 52$)
Balanced	B_1 ($n = 180$)	B_2 ($n = 277$)	B_3 ($n = 527$)
Ageing	A_1 ($n = 313$)	A_2 ($n = 114$)	A_3 ($n = 335$)

To assess partition stability under resampling [34], we re-fit both clustering stages on 50 random subsamples containing 85% of municipalities and compute the Adjusted Rand Index (ARI) [33] between subsample assignments and the full-sample reference. Demographic clustering (GMM, $n = 3$) achieves an ARI of 0.902 ± 0.043 , and socio-economic clustering within demographic groups (k-means, $n = 3$) achieves 0.959 ± 0.022 on average across the three groups. Both values indicate that cluster identity is well preserved under reasonable perturbations of the input sample, with the second stage being particularly stable. The lower variance in the second stage plausibly reflects the larger feature dimension (18 socio-economic indicators) and the smaller within-group sample sizes, which jointly produce sharper local optima.

Bootstrap ARI distributions are shown in Figure 6(a).



(a) Bootstrap ARI distributions over 50 subsampling runs for demographic clustering (GMM, $n = 3$) and within-group socio-economic clustering (k-means, $n = 3$, averaged across the three demographic groups) (b) Top-10 recommendation rank stability against the $k = 5$ baseline, measured by Jaccard overlap and Kendall's τ for $k \in \{3, 7, 10\}$. (c) Top-3 growth_gap and pressure topic stability against the 0.5 threshold baseline, for thresholds $\in \{0.3, 0.4, 0.6, 0.7\}$.

Figure 6. Stability and sensitivity analyses.

6.2. Recommendation robustness

The kNN recommendation mechanism has two principal degrees of freedom: the number of neighbors k and the within-cluster restriction on the candidate peer set. We assess sensitivity to both.

Sensitivity to k . Holding all other settings fixed, we vary $k \in \{3, 7, 10\}$ and measure the rank stability of the top-10 recommended factors against the $k = 5$ baseline using two metrics: Jaccard overlap of the top-10 sets and Kendall's τ on the full ranked feature list. We obtain Jaccard/ τ values of 0.74/0.52 at $k = 3$, 0.78/0.65 at $k = 7$, and 0.74/0.54 at $k = 10$. The non-monotone pattern is interpretable: very small neighborhoods produce noisier means, while very large ones dilute peer means with structurally less similar territories. The peak at $k = 7$ and the comparable value at $k = 5$ indicate that the default configuration sits in a stable region of the parameter space and that recommendation outputs are not driven by a particular choice of neighborhood size. Panel (b) of Figure 6 summarizes these values.

Within-cluster constraint ablation. To assess whether the cluster restriction is doing real work, we compare the constrained recommender (peer pool restricted to the same cluster) against an unconstrained variant in which the five nearest neighbors are drawn from the full set of 2,006 municipalities. Across all units, the average Jaccard overlap between the two top-10 lists is 0.815, and the mean Euclidean distance to chosen neighbors is 0.231 under the constraint and 0.202 without it. This pattern admits a coherent interpretation: for the majority of municipalities the unconstrained nearest neighbors already lie within the same cluster, so the constraint is not binding; for boundary cases the cluster restriction trades a modest proximity cost (about 14% larger neighbor distance) for drawing benchmarks from territories with comparable demographic and socio-economic structure. The 18.5% mean disagreement in top-10 recommendations is concentrated precisely in the units where the constraint matters most.

To illustrate this, we examine Peschanokopsky Municipal District, a case in the lower quartile of constrained–unconstrained agreement (Jaccard = 0.54; Figure 7). Under the cluster restriction all five peers belong to cluster A3 — Talovsky, Nizhnelomovsky, Tatsinsky, Povorinsky, and Khokholsky Municipal Districts. Without the restriction, only Talovsky and Nizhnelomovsky are retained, and the remaining peers come from territories with materially different demographic and socio-economic profiles (Semikarakorsky, Novopokrovsky,

and Veselovsky Municipal Districts). Of the resulting top-10 priority factors, seven are shared between the two variants. Three priorities — average salary, road length, and social-aid capacity — appear only under the cluster constraint. These are dimensions on which Peschanokopsky lags behind comparable peers but not behind the wider, structurally less similar pool, and would have been silently dropped by an unconstrained recommender. The contrast in mean neighbor distance is also informative (0.467 under the constraint versus 0.151 without it): the unconstrained version finds substantially closer matches in raw feature space, but those matches sit in different clusters, which is precisely the failure mode the typology is designed to prevent. This case illustrates the place-based motivation: the cluster constraint surfaces priorities calibrated to feasible peer benchmarks rather than to globally closest matches that may not be structurally comparable.

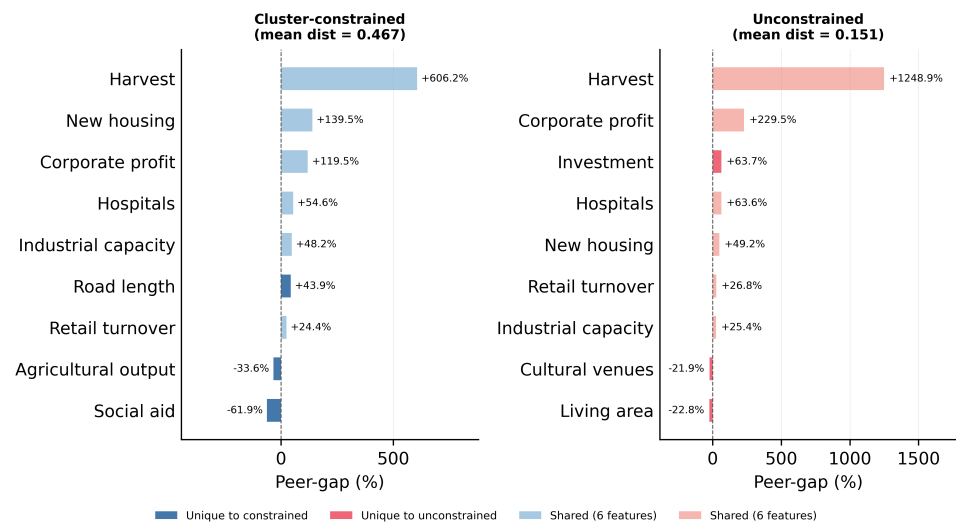


Figure 7. Within-cluster constraint ablation for Peschanokopsky Municipal District (lower-quartile case, Jaccard 0.54). Top-10 priority factors under the cluster-constrained recommender (left) and the unconstrained recommender (right); dark bars mark features unique to one variant, light bars the seven shared features. Under the constraint, all five peers are drawn from cluster A (mean neighbor distance 0.467); without it, only two of the original peers are retained, with closer matches in raw feature space (mean neighbor distance 0.151) drawn from different clusters.

6.3. Text pipeline validation

The BERTopic configuration (FRIDA embeddings, min_cluster_size = 50 and *n*-grams (1,2)) yields 1,119 topics, of which one is an outlier topic (-1) absorbing 208,738 of the 452,126 retained headlines. Topic coherence (c_v) is 0.387 on the full topic set and rises to 0.425 after the topic-filtering rule described in Section 5.3; topic diversity is 0.656. These values are consistent with reported BERTopic performance on short-text corpora. Zero-shot policy labeling (xlm-roberta-large-xnli, with the templates and labels described in Section 4.4) was evaluated on a manually annotated set of 461 headlines drawn at random from the filtered corpus. Overall accuracy is 0.597, macro- F_1 is 0.592, and weighted- F_1 is 0.594; per-class F_1 scores are 0.680 (progress), 0.537 (neutral), and 0.558 (crisis). The full confusion matrix (Figure 8) reveals a systematic asymmetry: the classifier over-predicts the crisis label (precision 0.44, recall 0.76) and under-predicts the neutral label (precision 0.67, recall 0.45), with neutral headlines distributed roughly evenly across the three predicted categories. We interpret this as evidence that crisis labels carry more label noise than progress labels in the downstream pipeline, and we accordingly treat the pressure score as a noisier signal than the growth-gap score. Crucially, we do not use the labels as ground truth at the headline level; they enter the framework only as inputs to topic–region aggregates, in which class-specific noise is partially averaged out.

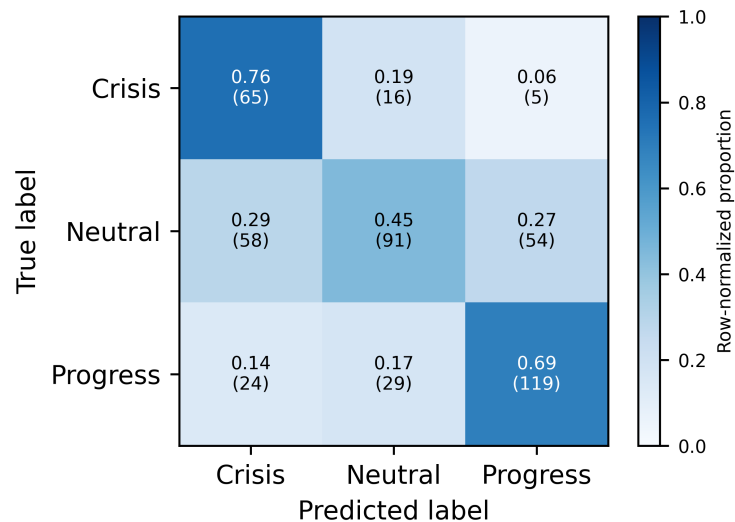


Figure 8. Row-normalized confusion matrix for the zero-shot classifier on the manually annotated evaluation set ($n = 461$). The over-prediction of *crisis* and under-prediction of *neutral* visible in the off-diagonal entries motivate cautious interpretation of the pressure score.

To evaluate sensitivity to the confidence threshold, we vary it in $\{0.3, 0.4, 0.6, 0.7\}$ and measure the mean overlap of the top-3 growth-gap and top-3 pressure topics against the 0.5 baseline. Lowering the threshold barely perturbs the rankings (overlap of 0.98 at both 0.3 and 0.4 for both signals), since few additional labels are accepted at low confidence levels. Raising the threshold tightens the signal but discards information: overlap drops to 0.79/0.68 at 0.6 and to 0.70/0.49 at 0.7 for growth-gap and pressure, respectively. Pressure top-3 lists are consistently more threshold-sensitive than growth top-3 lists, which is consistent with the over-prediction of crisis visible in the confusion matrix. The default threshold of 0.5 sits at the elbow of the stability curve and balances signal retention against label noise. Panel (c) of Figure 6 reports the corresponding overlap values.

7. Discussion and Limitations

This study presents a practical framework for place-based municipal revitalization recommendations that explicitly respects territorial heterogeneity. The two-stage typology provides a defensible comparability principle: municipalities are benchmarked against peers with similar demographic structure and socio-economic profiles. This design reduces the risk of unrealistic policy targets that arise when small or rural municipalities are compared directly with metropolitan areas. The recommendation mechanism is intentionally simple and transparent—cluster-constrained kNN gaps—making it suitable for decision-support settings where interpretability and traceability are required.

The integration of regional news adds a contextual layer that may help explain or complement numeric gaps. Topic-based growth_gap and pressure indicators provide a structured way to compare how "progress" and "crisis" narratives differ between a municipality's region and peer regions. However, these signals should be interpreted cautiously. First, news is only observed at the regional level, and the mapping to municipalities via news-per-population is an approximation that cannot capture within-region media heterogeneity. Second, zero-shot classification shows moderate performance (accuracy 0.597; macro- F_1 0.592), implying label noise [25] and potential systematic biases by topic or region. Third, the analysis is cross-sectional (2022 only), so recommendations are correlational and cannot establish causal effects on migration outcomes.

More broadly, the framework illustrates a pattern likely to recur in smart-city decision support: structural and textual data streams operate at different spatial granularities, and

integrating them requires explicit assumptions about how coarser signals map to the unit of decision. Making those assumptions transparent — as we do via the news-per-population weight — is, in our view, a precondition for trustworthy data-driven urban governance.

Despite these limitations, the framework is useful as a structured benchmarking tool [2,3]. It can highlight plausible priorities, support exploratory analysis, and generate hypotheses for deeper policy investigation. The proposed validation plan outlines steps to strengthen robustness, including stability analysis, sensitivity to k and thresholds, and outcome-oriented simulation once longitudinal data become available.

8. Conclusion and Future Work

We developed a municipality-level framework for place-based revitalization recommendations that combines (i) a two-stage typology (demographic clustering followed by socio-economic clustering within demographic groups), (ii) transparent numeric recommendations via cluster-constrained kNN gap ranking, and (iii) a contextual news layer using semantic filtering, BERTopic topic modeling, and zero-shot policy labeling. Applied to 2,006 municipalities and 2022 regional news headlines, the approach produces interpretable municipality profiles and configurable recommendation outputs (top-10 numeric factors and top-3 progress/crisis topics by default). Two case studies — Kamensky Municipal District (Section 5.2) and Peschanokopsky Municipal District (Section 6.2) — illustrate the framework's concrete outputs and demonstrate that the cluster constraint materially shapes the recommended priorities for municipalities at the boundary of their typology cell.

Future work will focus on (1) strengthening the text component by improving labeling quality (e.g., domain-adaptive fine-tuning or active learning), (2) refining the municipality-news linkage beyond regional aggregation when finer-grained media data become available, (3) conducting systematic robustness and stability evaluations, and (4) moving toward outcome-oriented validation using longitudinal municipal indicators and migration dynamics. Ultimately, the framework aims to support analysts and policymakers in forming more realistic peer benchmarks and prioritizing interventions aligned with both structural gaps and contextual signals.

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